IIOT External Viva

Question & Answer Booklet | Computer Laboratory II | Course Code 417526

Practical-wise viva questions with short, easy-to-speak answers, plus common IIoT fundamentals.

Preparation tip: Read each question aloud and practise answering in one or two sentences. For practical-specific questions, use the pin numbers and observations stated in your lab manual.

Source note: The practical-specific questions and answers are based on your uploaded IIOT lab manual (experiments 1–5, pages 11–25). Where the manual does not specify a detail, this booklet avoids presenting it as a manual requirement.

Practical 1 – Potentiometer with ESP32

Q1. What is a potentiometer?

Answer: A potentiometer is a variable resistor used to vary resistance and produce a changing output voltage.

Q2. What is ESP32?

Answer: ESP32 is a microcontroller board with built-in Wi-Fi and Bluetooth capabilities.

Q3. What is GPIO?

Answer: GPIO means General Purpose Input/Output. These pins connect sensors and other devices to a microcontroller.

Q4. What is the function of the potentiometer in this practical?

Answer: It provides a variable analog voltage that ESP32 reads to generate a warning when the reading crosses a set threshold.

Q5. Which ESP32 pin is connected to the potentiometer?

Answer: GPIO 34.

Q6. Why is GPIO 34 used?

Answer: GPIO 34 supports analog input and is suitable for reading the potentiometer voltage.

Q7. What is ADC?

Answer: ADC means Analog-to-Digital Converter. It converts an analog signal into a digital value.

Q8. What is the analogRead() range used in this practical?

Answer: The manual describes a 12-bit ADC reading range of 0 to 4095.

Q9. Why is the maximum reading 4095?

Answer: A 12-bit ADC has 2^{12} = 4096 possible values, numbered from 0 to 4095.

Q10. At what value is the warning generated?

Answer: When the potentiometer reading is greater than 2000.

Q11. What is Serial Monitor?

Answer: It is an Arduino IDE tool used to display program output and debugging messages.

Q12. What is baud rate?

Answer: It is the communication speed setting for serial communication. This practical uses 115200.

Q13. What is the purpose of delay(500)?

Answer: It pauses the program for 500 milliseconds, or half a second.

Q14. What is the difference between setup() and loop()?

Answer: setup() runs once when the board starts; loop() runs repeatedly.

Q15. What happens if the reading is 1500?

Answer: The reading is displayed, but the warning condition is not met.

Q16. What happens if the reading is 3000?

Answer: The reading is displayed and the warning message is printed.

Q17. What is the difference between analogRead() and digitalRead()?

Answer: analogRead() reads an analog input as a numerical value; digitalRead() reads a digital state such as HIGH or LOW.

Q18. Why should GPIO 34 not be used to drive an LED?

Answer: GPIO 34 is an input-only pin on ESP32; it is not suitable as a digital output.

Practical 2 – PIR Sensor with Arduino

Q1. What is a PIR sensor?

Answer: PIR stands for Passive Infrared Sensor. It detects motion by sensing changes in infrared radiation.

Q2. What is the full form of PIR?

Answer: Passive Infrared.

Q3. Why is it called passive?

Answer: It does not emit infrared radiation; it detects infrared radiation from its surroundings.

Q4. What is the working principle of a PIR sensor?

Answer: It detects changes in infrared radiation when a warm object, such as a person, moves within its detection area.

Q5. Where are PIR sensors used?

Answer: They are used in security alarms, automatic lighting, motion detection and smart-home systems.

Q6. How many pins does a typical PIR sensor have?

Answer: Three: VCC, GND and OUT.

Q7. Which Arduino pin receives the PIR output in this practical?

Answer: Digital pin 2.

Q8. What is the output of a PIR sensor?

Answer: It gives a digital output: HIGH when motion is detected and LOW when no motion is detected.

Q9. What is digitalRead()?

Answer: It reads the digital state of a pin as HIGH or LOW.

Q10. What is pinMode()?

Answer: It configures a microcontroller pin as INPUT or OUTPUT.

Q11. Why is Serial.begin(9600) used?

Answer: It starts serial communication at 9600 baud so messages can be viewed in Serial Monitor.

Q12. What happens when motion is detected?

Answer: The Arduino reads HIGH, turns the LED ON and prints a motion-detected message.

Q13. What happens when no motion is detected?

Answer: The Arduino reads LOW, turns the LED OFF and prints a no-motion message.

Q14. Can a PIR sensor continuously detect a stationary person?

Answer: It mainly detects changes in infrared radiation caused by movement, so it may not continuously detect a stationary person.

Q15. Why is an LED used in this experiment?

Answer: It gives a visible indication of whether motion has been detected.

Q16. What is Tinkercad?

Answer: Tinkercad is an online platform that can be used to design and simulate electronic circuits.

Q17. What is the difference between PIR and an active IR sensor?

Answer: A PIR sensor detects changes in infrared radiation; an active IR sensor emits infrared light and detects its reflection or interruption.

Practical 3 – CPS Monitoring and Control

Q1. What is CPS?

Answer: CPS stands for Cyber-Physical System. It integrates physical processes with computer-based monitoring and control.

Q2. Give examples of CPS.

Answer: Industrial automation, autonomous vehicles, smart cities and smart manufacturing.

Q3. What is the aim of this practical?

Answer: To create a web interface that allows a user to monitor and control LEDs connected to ESP32 over Wi-Fi.

Q4. What is a web server?

Answer: A web server receives requests from clients and sends responses such as webpages.

Q5. What is an ESP32 web server?

Answer: It is a web server hosted on the ESP32 that can be accessed through a browser over a network.

Q6. Which communication technology is used?

Answer: Wi-Fi.

Q7. What is an IP address?

Answer: An IP address is a logical network address used to identify a device on a network.

Q8. How do we find the ESP32 IP address?

Answer: After ESP32 connects to Wi-Fi, its local IP address is printed in the Arduino Serial Monitor.

Q9. Which GPIO pins are used for the LEDs in the manual?

Answer: GPIO 12 and GPIO 14.

Q10. Why is a 220-ohm resistor used?

Answer: It limits current through the LED and helps protect it from damage.

Q11. What is HTML?

Answer: HTML stands for HyperText Markup Language. It defines the structure of a webpage.

Q12. What is HTTP?

Answer: HTTP stands for HyperText Transfer Protocol. It is used for communication between web clients and servers.

Q13. What happens when the ON control is clicked?

Answer: The browser sends a request to ESP32, which sets the selected GPIO pin HIGH and turns the LED ON.

Q14. What happens when OFF is clicked?

Answer: ESP32 sets the selected GPIO pin LOW and turns the LED OFF.

Q15. What is monitoring?

Answer: Monitoring means observing the current status of a system.

Q16. What is controlling?

Answer: Controlling means sending commands to change how a system operates.

Q17. Can a mobile phone access the ESP32 webpage?

Answer: Yes, if it is connected to the same accessible network as the ESP32.

Q18. What happens if the IP address is incorrect?

Answer: The browser will not reach the ESP32 web server.

Q19. What is the difference between ESP32 and standard Arduino Uno?

Answer: ESP32 has built-in Wi-Fi and Bluetooth; the standard Arduino Uno does not.

Practical 4 – Sensor Data on a Webpage

Q1. What is the aim of this practical?

Answer: To read potentiometer data using ESP32 and display the reading on a webpage hosted by ESP32.

Q2. Which sensor is used?

Answer: A 10K potentiometer.

Q3. Which pin reads the potentiometer?

Answer: GPIO 34.

Q4. Which library provides Wi-Fi functionality?

Answer: WiFi.h.

Q5. Which library provides web-server functionality?

Answer: WebServer.h.

Q6. What does WebServer server(80) mean?

Answer: It creates a web-server object that listens for HTTP requests on port 80.

Q7. What is port 80?

Answer: It is the standard default port for HTTP communication.

Q8. What is SSID?

Answer: SSID stands for Service Set Identifier; it is the name of a Wi-Fi network.

Q9. What does WiFi.begin() do?

Answer: It starts connecting ESP32 to the Wi-Fi network using the supplied network name and password.

Q10. What is handleRoot()?

Answer: It is a function that handles the root webpage request and prepares the page containing the sensor reading.

Q11. What does server.on() do?

Answer: It registers a URL route and the function to run when that route is requested.

Q12. What is HTTP_GET?

Answer: It is an HTTP method used by a client to request information from a server.

Q13. What does server.begin() do?

Answer: It starts the web server so it can accept requests.

Q14. What does server.handleClient() do?

Answer: It checks for and processes incoming client requests.

Q15. What is the role of analogRead()?

Answer: It reads the analog voltage at GPIO 34 and returns a digital reading.

Q16. What is the difference between WiFi.h and WebServer.h?

Answer: WiFi.h connects ESP32 to a wireless network; WebServer.h lets ESP32 handle HTTP requests and serve webpages.

Q17. Does this practical store data in an external cloud database?

Answer: No. The implementation described in the lab manual hosts a local webpage on ESP32 and displays the potentiometer reading; it does not describe external cloud-database storage.

Q18. What happens if Wi-Fi disconnects?

Answer: The browser cannot access the ESP32 webpage over that network until connectivity is restored.

Q19. What is the difference between HTTP and HTTPS?

Answer: HTTP does not provide built-in encryption; HTTPS uses TLS to encrypt communication.

Practical 5 – Industrial Data Analysis using Excel

Q1. What is data analytics?

Answer: It is the process of collecting, processing and analyzing data to find useful information and support decisions.

Q2. What is IoT data analytics?

Answer: It is the analysis of data generated by connected IoT devices and sensors to identify patterns and useful insights.

Q3. What is the aim of this practical?

Answer: To analyze farm temperature and humidity readings using Excel charts and graphs.

Q4. Which software is used?

Answer: Microsoft Excel.

Q5. Where was the dataset obtained?

Answer: Kaggle.

Q6. What is the dataset name?

Answer: Farm Temperature and Humidity.

Q7. What is Kaggle?

Answer: Kaggle is an online platform that provides datasets, data-science competitions and learning resources.

Q8. Which parameters are analyzed?

Answer: Temperature and humidity.

Q9. How many farms are included?

Answer: Four farms, approximately one kilometre apart.

Q10. What is the duration of the dataset?

Answer: Three months.

Q11. How often were readings taken?

Answer: Every one hour.

Q12. What is data visualization?

Answer: It is the graphical representation of data using charts, graphs and dashboards.

Q13. Why are graphs used?

Answer: They make patterns, trends and variations easier to understand.

Q14. What is a line chart?

Answer: A chart that connects data points with lines and is useful for showing changes over time.

Q15. What is a scatter plot?

Answer: A graph that displays individual data points to show distributions, patterns or possible relationships.

Q16. What is the difference between temperature and humidity?

Answer: Temperature measures how hot or cold something is; humidity measures the amount of water vapour in the air.

Q17. What is data cleaning?

Answer: It is the process of identifying and correcting missing, duplicate, inconsistent or incorrect data.

Q18. What is data processing?

Answer: It is the transformation of raw data into a useful format for analysis.

Q19. What components of IoT analytics are listed in the manual?

Answer: Data collection, data storage, data processing and data analysis; the manual also describes data visualization.

Q20. What is the 24-hour temperature observation?

Answer: The manual reports a temperature peak around 12 o'clock, followed by a decrease later and at night.

Q21. What is the 48-hour temperature observation?

Answer: The manual reports peaks around hour 12 and hour 36.

Q22. What happens to humidity at night according to the manual?

Answer: The manual's observation says humidity increases at night.

Q23. What is observed around hours 13 and 37 in the humidity graph?

Answer: The manual reports a similar fall in humidity around those times.

Q24. What is predictive analytics?

Answer: It uses historical data and statistical or machine-learning methods to estimate future outcomes.

Q25. How is data analytics useful in industries?

Answer: It can help monitor equipment, identify trends, improve productivity and support decisions.

Common IIoT Viva Questions

Q1. What is IoT?

Answer: Internet of Things: a network of connected physical devices that collect and exchange data.

Q2. What is IIoT?

Answer: Industrial Internet of Things: connected sensors, machines and systems used in industrial environments.

Q3. What is the difference between IoT and IIoT?

Answer: IoT commonly covers consumer and everyday applications; IIoT focuses on industrial monitoring, automation, reliability and productivity.

Q4. What is a sensor?

Answer: A device that detects a physical parameter such as temperature, pressure, humidity or motion.

Q5. What is an actuator?

Answer: A device that converts a control signal into a physical action, such as movement or switching.

Q6. What is the difference between a sensor and an actuator?

Answer: A sensor measures or detects; an actuator performs an action.

Q7. What are communication protocols?

Answer: Rules that allow devices to exchange data. Examples include HTTP and MQTT.

Q8. What is MQTT?

Answer: Message Queuing Telemetry Transport, a lightweight messaging protocol commonly used in IoT.

Q9. What is cloud computing?

Answer: Delivery of computing services such as storage, servers and processing over a network, commonly the internet.

Q10. What is edge computing?

Answer: Processing data close to where it is generated instead of sending all data to a distant cloud.

Q11. What is a Digital Twin?

Answer: A virtual representation of a physical object or system that uses real-world data for monitoring and analysis.

Q12. What is predictive maintenance?

Answer: Using equipment and sensor data to identify possible failures before they happen.

Q13. What are benefits of IIoT?

Answer: Real-time monitoring, improved productivity, reduced downtime, better maintenance planning and data-based decisions.

Q14. What are IIoT security risks?

Answer: Unauthorized access, weak passwords, malware, data breaches and insecure networks.

Q15. What is Arduino IDE?

Answer: An Integrated Development Environment used to write, compile and upload programs to supported microcontroller boards.

Last-Minute Memory Sheet

Practical	Key facts to remember
1 – Potentiometer	ESP32 GPIO 34; 3.3V and GND; ADC 0–4095; warning when reading > 2000; Serial 115200.
2 – PIR	PIR OUT to Arduino D2; VCC to 5V; GND to GND; HIGH = motion; LOW = no motion; Serial 9600.
3 – CPS Web Control	ESP32 LEDs on GPIO 12 and 14; 220Ω resistors; Wi-Fi web server; HTTP port 80; Serial 115200.
4 – Sensor Webpage	Potentiometer to GPIO 34; WiFi.h and WebServer.h; server port 80; local webpage shows sensor reading.
5 – Excel Analysis	Kaggle Farm Temperature and Humidity dataset; 4 farms; 3 months; readings hourly; line charts and scatter plots.

Exam strategy: For every experiment, be ready to state the aim, name the components, explain the connections, describe the main code function and tell the examiner the expected output.

Prepared from the uploaded Computer Laboratory II: Industrial Internet of Things manual, Course 417526. Answers are intentionally concise for viva revision.